

AI LEGAL™

Quiz & MCQ Practice Module

Comprehensive Product, Technical & Architecture Documentation

Document Area	Details
Module Route	/dashboard/tools/quiz-practice
Primary Experience	Topic Selection → Active Test → Scorecard & Diagnostics
Core Concept	AI-Powered Legal Examination Preparation & Self-Assessment Engine
Target Users	Law Students, AIBE Candidates, CLAT UG/PG Aspirants & PCS-J Aspirants

Document Structure

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1. Executive Summary & Module Purpose

The Quiz & MCQ Practice Module (/dashboard/tools/quiz-practice) is an advanced AI-powered examination preparation and self-assessment engine designed for law students, All India Bar Examination (AIBE) candidates, CLAT UG/PG aspirants, and State Judicial Services (PCS-J) aspirants.

The module combines curated high-yield question banks across 30+ legal subjects with an on-demand AI MCQ generation pipeline capable of producing exam-standard multiple-choice questions, statutory scenario dilemmas, and case-law application questions.

Each question is designed to provide immediate or deferred pedagogical feedback, including answer rationale, statutory references, relevant case-law context, progressive hints, and performance-oriented remediation.

2. Core Value Propositions

Value Proposition	Description
Targeted Exam Simulation	Exam-oriented environments for State Judiciary Prelims (PCS-J), AIBE, CLAT UG/PG, and Nepal Bar Council / Judicial Service preparation.
Dual Question Architecture	Combines curated high-yield question banks with dynamically generated AI MCQs for doctrines, judgments, statutes, and recent legal developments.
Deep Explanatory Diagnostics	Every question can provide granular reasoning explaining the correct answer, distractor failures, statutory provisions, and relevant case-law principles.
Progressive Hint Engine	Multi-tier hints support legal reasoning and deductive statutory interpretation without immediately revealing the answer.
Comprehensive Performance Analytics	Post-quiz diagnostics measure accuracy, speed, question-level performance, topic mastery, and remediation opportunities.

3. Subject Catalog & Jurisdictional Coverage

The module is designed around a multi-jurisdictional subject catalog. Indian legal preparation forms the primary 30+ subject library, complemented by Nepal-focused legal and examination content.

3.1 Indian Jurisdictional Catalog

- Criminal Jurisprudence: BNS 2023, BNSS 2023, BSA 2023, IPC 1860, CrPC 1973 — including substantive provisions, procedure, evidence, bail, general exceptions, mens rea, electronic evidence, and modern criminal-law reforms.
- Constitutional Law: Fundamental Rights (Articles 12–35), Directive Principles and Fundamental Duties, writ jurisdictions, higher judiciary, and major constitutional amendments including the 42nd, 44th, 86th, 99th, 101st, 103rd, and 106th Amendments.
- Civil & Commercial Jurisprudence: Indian Contract Act, CPC, Specific Relief Act, Limitation Act, res judicata, injunctions, rejection of plaint, specific performance, limitation and delay principles.
- Corporate, Cyber & Specialized Statutes: Companies Act, IBC, Information Technology Act, DPDP framework, Negotiable Instruments Act Section 138, Copyright, Trademark, and Patent fundamentals.

3.2 Nepal Jurisdictional Catalog

- Constitution of Nepal 2072: Fundamental rights and extraordinary writ jurisdictions.
- Muluki Criminal Code 2074 & Procedure: Homicide, fraud (फ्रॉड), bail/hearing concepts (बैल/सुनवाई), FIR/complaint processes (फाइल/शिकायत), and criminal procedure.

- Muluki Civil Code 2074 & Procedure: Contractual obligations, torts, interim injunctions, and limitation (संवैधानिक).
- Special Statutes: Evidence Act 2031, Banking Offence Act 2064, Electronic Transactions Act 2063.
- Exam Specials: Nepal Bar Council Advocate Examination and Judicial Service Examination preparation.

4. Quiz Workflow & Interaction Lifecycle

Stage	Primary Actions
1. Question Source	Curated Topic Selection OR AI Topic / Query Generation.
2. Active Test Environment	Timed or untimed mode, option selection, feedback behaviour, progressive hints, and question bookmarking.
3. Submission	Practice mode can provide immediate feedback; Mock Exam mode submits the full test as a batch.
4. Scorecard & Diagnostics	Score, accuracy, speed, detailed explanations, incorrect-question filtering, and retake/remediation actions.

4.1 Mode Selection & Setup

- Practice Mode: Immediate feedback on option selection, intended for daily statutory and concept study.
- Mock Exam Mode: Countdown-based environment in which questions are submitted as a complete test and feedback is withheld until final submission.

4.2 In-Quiz Interactive Experience

- Option Shuffling: Randomizes option positions to reduce positional bias.
- Instant Option Validation: Practice Mode provides immediate correct/incorrect visual feedback.
- Pedagogical Explanation Drawer: Shows statutory provisions, reasoning, and relevant case-law principles supporting the answer.
- Deductive Hint Tool: Provides contextual guidance without directly naming the correct option.
- Bookmark Question: Allows users to save difficult or high-value questions for later revision.

4.3 Performance Scorecard & Remediation

- Composite Metrics: Total questions answered, correct/incorrect count, accuracy percentage, and average time per question.
- Retake Missed Questions Only: Creates a focused retry set from incorrect responses.
- Export to AI Notes Workspace: Sends difficult questions and explanations into the student's personal notes.
- Bookmark Catalog: Stores tagged questions inside a persistent revision vault.

5. Technical Implementation Details

Layer	Implementation	Purpose
Web Frontend	QuizPracticeWorkspace.jsx	Primary Quiz & MCQ workspace for web.
Mobile Frontend	quiz-practice.tsx	Mobile implementation of quiz practice.
AI Generation	Google Gemini API / generateChatResponse	Dynamic MCQ generation using structured prompts and JSON constraints.
Question Schema	Structured 4-option output	Enforces valid option sets, answer keys, explanations, and statutory citations.

Layer	Implementation	Purpose
Quiz History	student_quiz_history_	Persists user quiz activity and historical attempts.
Bookmarks	student_quiz_bookmarks_	Persists bookmarked questions for revision.
Study Streak	recordStudyActivity	Completing a quiz updates the student's daily study streak.

6. Cross-Module Synergies

- AI Knowledge Hub: Launch topic-specific quizzes directly while reading a bare-act section.
- AI Notes Maker: Convert quiz errors, explanations, and difficult concepts into formatted study notes or flashcards.
- Student Learning Portal: Quiz completion feeds into Today's Challenge and the overall Study Progress percentage.
- Future Research Integration: Questions can link back to statutory provisions, legal precedents, and knowledge-hub sections for deeper revision.

7. Functional Requirements & UX Behaviour

Requirement	Expected Behaviour
Question Quality	Curated questions should be reviewed for legal accuracy; AI-generated questions should follow schema validation and citation requirements.
Exam Modes	Practice and Mock Exam modes must behave distinctly with respect to feedback, timing, and submission.
Timer	Mock mode should support a visible countdown and preserve the current attempt state.
Feedback	Practice mode should show answer validation and explanation without disrupting question flow.
Hints	Hints should be progressively revealed and avoid directly disclosing the correct option.
Analytics	Scorecard should clearly expose accuracy, speed, topic-level performance, and missed-question remediation.
Persistence	History and bookmarks should remain available across sessions for the same user account.
Accessibility	Controls, answer states, timers, and explanations should remain readable and usable across desktop and mobile layouts.

8. Data, Persistence & Performance Model

The module follows a user-centric quiz state model in which each attempt can be represented as:

User → Exam/Mode → Topic → Question Set → Question State → Answer/Hints → Submission → Scorecard → Remediation

Persistent storage separates historical attempts from bookmarked revision content. This allows the product to support repeated testing, targeted retry sets, study streak tracking, and longitudinal performance analysis.

9. Future Extensions

- Adaptive difficulty that adjusts question complexity based on recent performance.
- Subject-wise mastery heatmaps and weak-topic detection.
- Exam-specific full-length mock tests with configurable negative marking and section rules where applicable.
- AI-generated case-law and judgment application questions using verified legal sources.
- Spaced-repetition scheduling for bookmarked and repeatedly missed questions.
- Leaderboards or cohort analytics where enabled by product policy and user consent.
- Offline-capable mobile practice with synchronization when connectivity is restored.

Appendix A — Example User Journey

- 1. User opens Quiz & MCQ Practice.
- 2. User chooses an exam, jurisdiction, subject, and question count.
- 3. User selects Practice Mode or Mock Exam Mode.
- 4. System loads curated questions or generates a validated AI question set.
- 5. User answers questions, uses hints, and bookmarks difficult items.
- 6. User submits the test.
- 7. System displays score, accuracy, timing, explanations, and topic-level diagnostics.
- 8. User retakes missed questions or exports difficult material to AI Notes.
- 9. Completion updates the Student Learning Portal and study streak.

Appendix B — Reference Implementation Paths

Area	Reference
Web	AI-Legal_App_Webapp/src/pages/QuizPracticeWorkspace.jsx
Mobile	AI-Legal_App_frontend-mobile/src/app/tools/quiz-practice.tsx
AI Engine	generateChatResponse / Google Gemini integration
History	student_quiz_history_
Bookmarks	student_quiz_bookmarks_
Streak	recordStudyActivity